Computer Networks I

SUMITA TYAGI

What is Computer Networks?

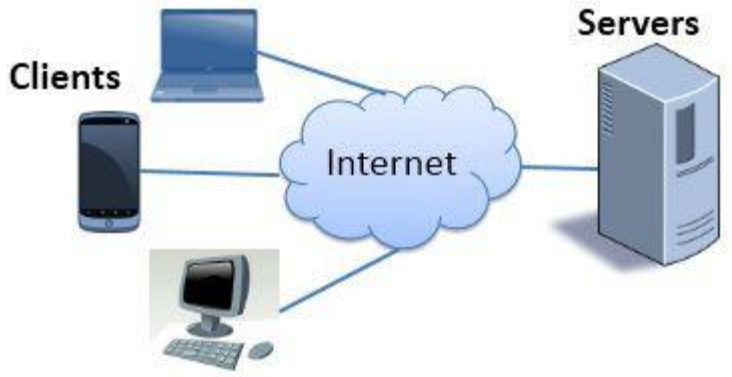
- **Computer network** A collection of computing devices that are connected in various ways to communicate and share resources

  Usually, the connections between computers in a network are made using physical wires or cables

  However, some connections are wireless, using radio waves or infrared signals

Client Server Architecture

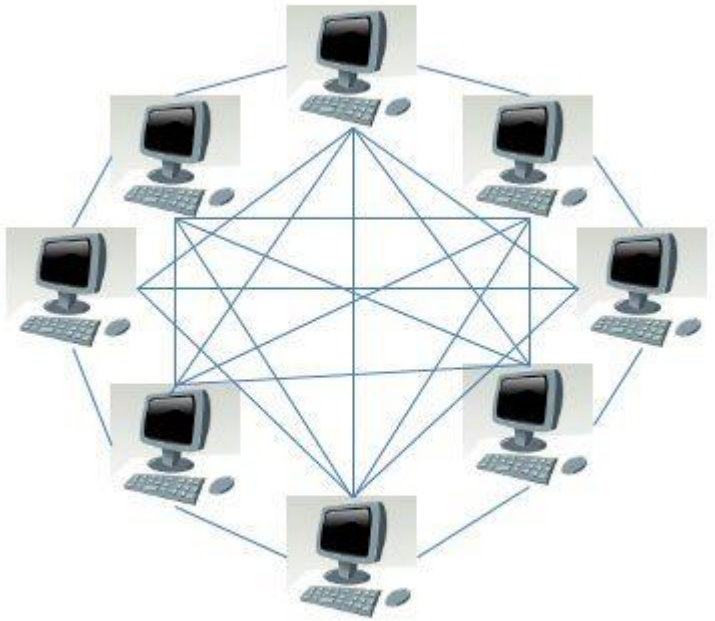
- The Client-Server network model is a widely used network model. Here, **Server** is a powerful system that stores the data or information in it. On the other hand, the **Client** is the machine that lets users access the data on the remote server. It's always the client requesting a service from the server, and the server is responsible for providing the service.



Client-Servers Network Model

Peer to peer architecture

- Unlike the client-server model, the Peer-to-Peer model does not distinguish between client and server; instead, each node can be either a client or a server, depending on whether it is **requesting** or **providing** services. Each node is considered a **peer**.



Peer-to-Peer Network Model

BASIS FOR COMAPAISON	CLIENT-SERVER	PEER-TO-PEER
Basic	There is a specific server and specific clients connected to the server.	Clients and server are not distinguished; each node act as client and server.
Service	The client request for service and server respond with the service.	Each node can request for services and can also provide the services.
Focus	Sharing the information.	Connectivity.
Data	The data is stored in a centralized server.	Each peer has its own data.
Server	When several clients request for the services simultaneously, a server can get bottlenecked.	As the services are provided by several servers distributed in the peer-to-peer system, a server in not bottlenecked.
Expense	The client-server are expensive to implement.	Peer-to-peer are less expensive to implement.
Stability	Client-Server is more stable and scalable.	Peer-to-Peer suffers if the number of peers increases in the system.

LAN, MAN and WAN

- LAN covers a small geographical area (up to 1km), whereas MAN covers a relatively large area (5 to 50 km), and WAN covers the largest area of all.
- LAN is confined to schools, hospitals or buildings, whereas MAN connects small towns or cities, and WAN covers a country or a group of Countries.
- Devices used for transmission of data are-
  LAN: WiFi, Ethernet Cables.
  MAN: Modem and Wire/Cable
  WAN: Optical wires, Microwaves, Satellites.
- LAN's transmit data at a faster rate than MAN and WAN.
- The bandwidth available for transmission is higher in LAN than in MAN and WAN.
- Data transmission errors and noise are least in LAN, moderate in MAN and high in WAN.

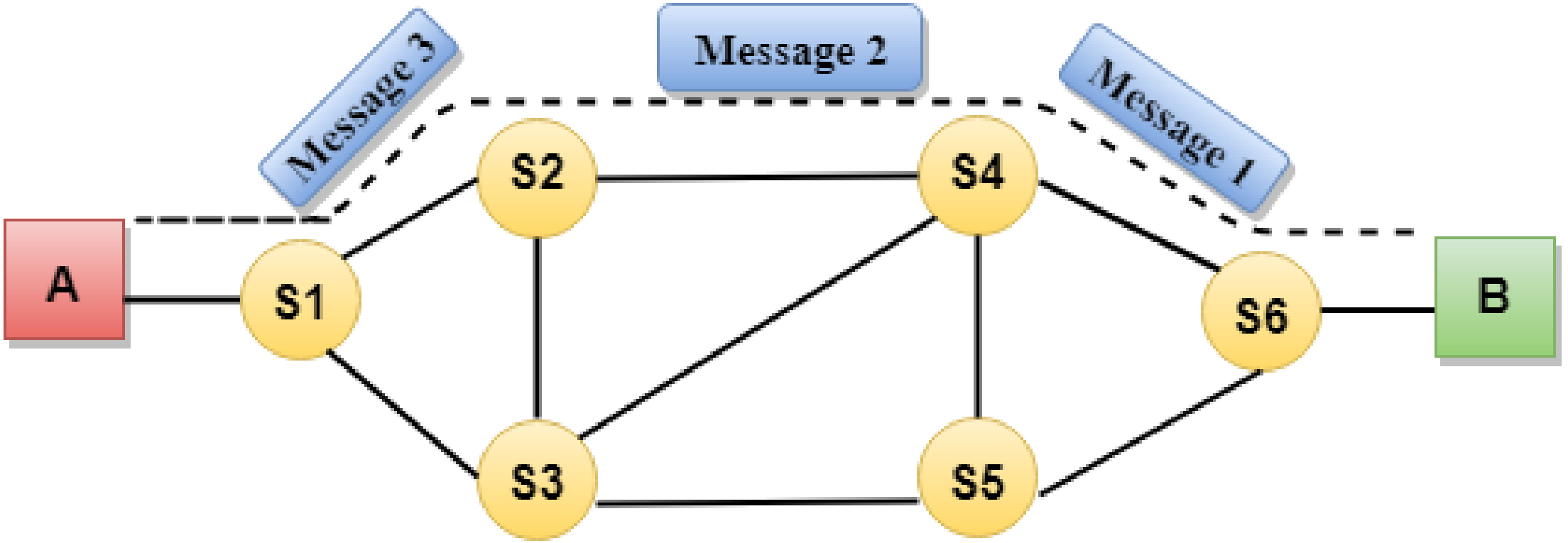
PAN

- A personal area network (**PAN**) is the interconnection devices within the **range** of an individual person, typically within a **range** of 10 meters.

Circuit Switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- Circuit switching in a network operates in a similar way as the telephone works.
- A complete end-to-end path must exist before the communication takes place.
- Fixed data can be transferred at a time in circuit switching technology. There is no limit on the data.

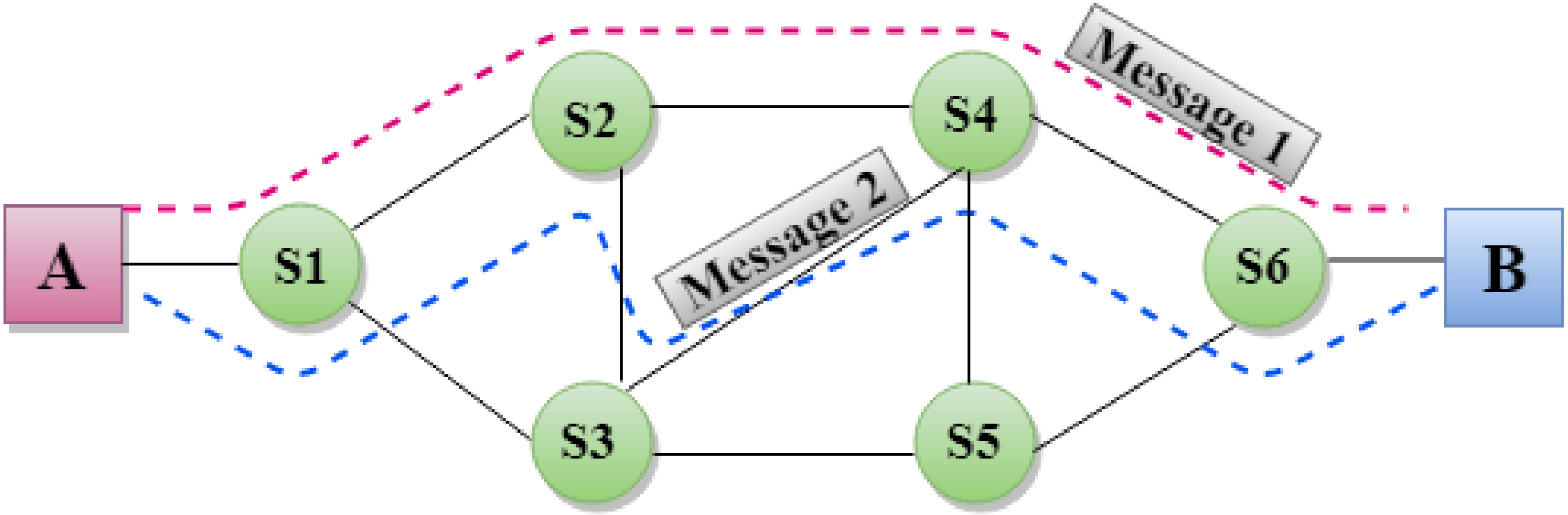
Circuit Switching



Message Switching

- Message Switching is a switching technique in which a message is routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as **store and forward network.**
- Data is divided into small messages

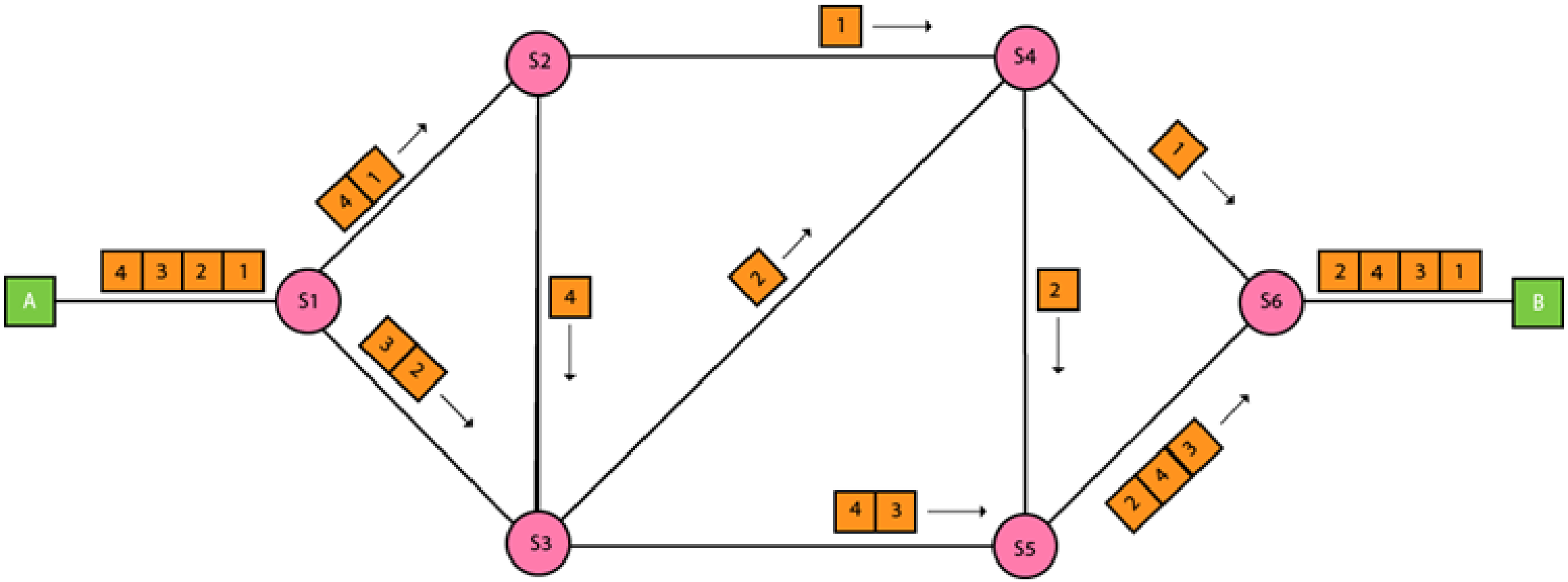
Message Switching



Packet Switching

- The packet switching is a switching technique in which the message is divided into equal sized pieces known as packets, and they are sent individually.
- Each packet is given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.

Packet Switching

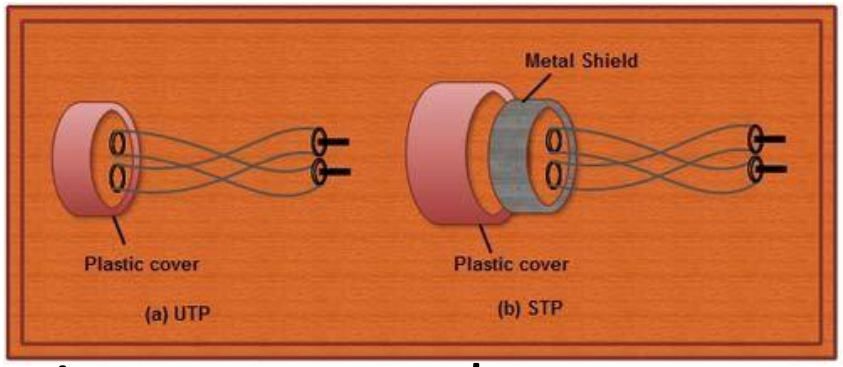


Wired Media(Twisted Pair)

It is made up of two copper wires twisted together. Types of twisted pairs are:

STP(Shielded Twisted Pair)

UTP(Unshielded Twister Pair)



Why twist the wires?

- Twisting of wires will reduce the effect of noise or external interference.
- The number of twists per unit length will determine the quality of the cable.

More twists mean better quality.

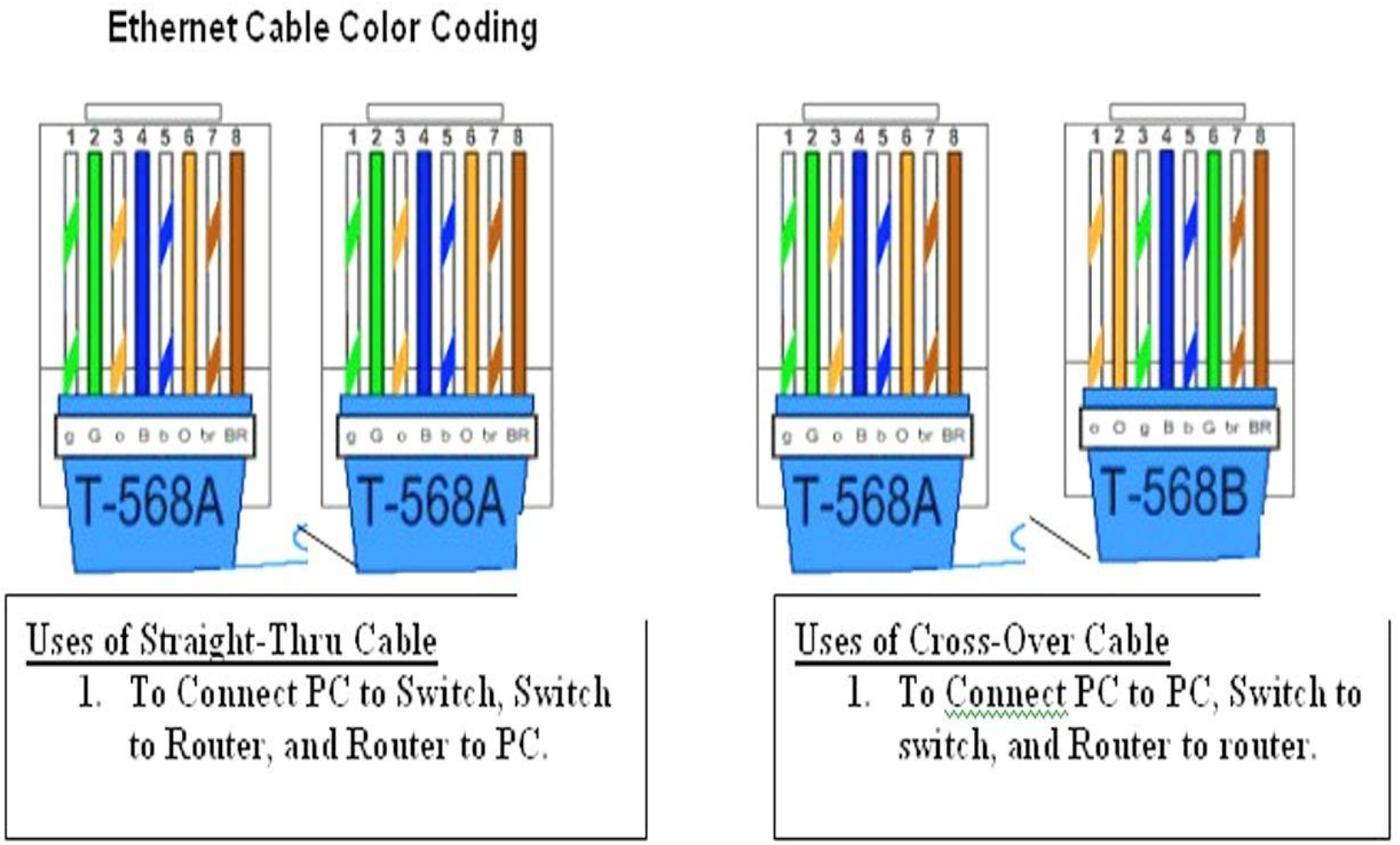
Wired Media(Twisted Pair)

- **USES:**

In telephone cables and internet connections

- **Advantages of twisted-pair cable**
- It can be used to carry both analogue and digital data.
- It is relatively easy to implement and terminate.
- It is the least expensive medium of transmission for short distances.
- If a portion of a twisted pair cable is damaged, it does not affect the entire network.
- **Disadvantages of Twisted pair cable**
- It offers poor noise immunity; as a result, signal distortion is greater.
- Attenuation is very high.
- It offers very poor security and is relatively easy to tap.
- Being thin in size, they are likely to break easily.

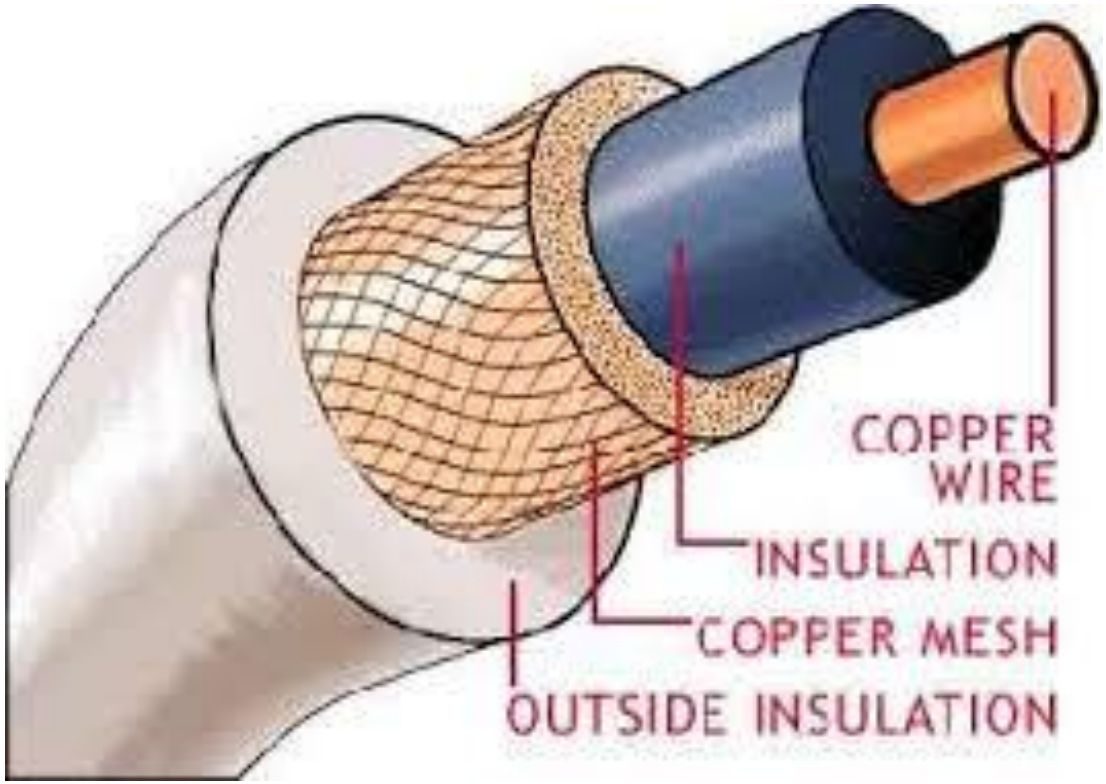
Colour coding in twisted pair



Connector for Twisted Pair: RJ(Registered Jack)



Wired Media(Coaxial Cable)



Used basically for TV connections

Connector: BNC(Bayonet Neill–Concelman)



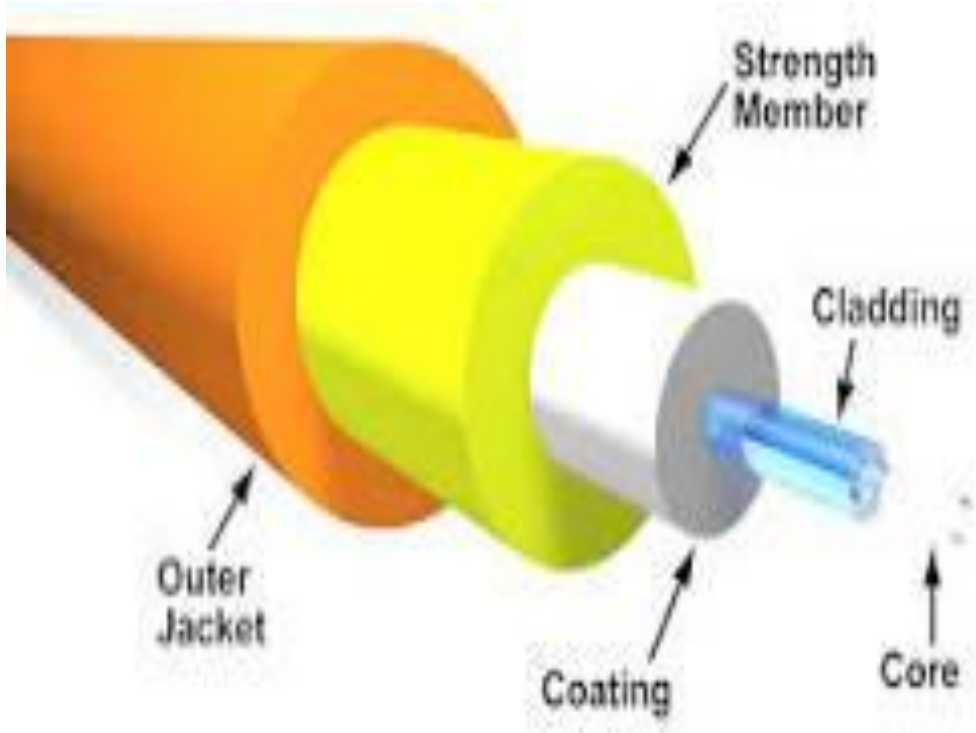
Advantages and Disadvantages of coaxial cable

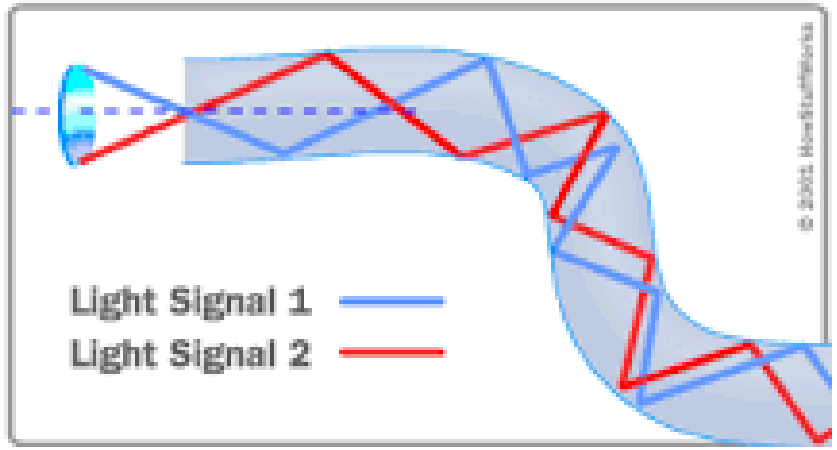
- **Advantages**
- Sufficient frequency range to support multiple channels, which allows for much greater throughput.
- Lower error rates
- Greater spacing between amplifiers and coax cable shielding reduces noise and crosstalk.

Disadvantages

- More expensive to install compared to twisted-pair cable.
- The thicker the cable, the more difficult it is to work with.

Wired Media(Optical Fibres)





Core is made up of very thin fibre of silica or glass or plastic. The core is then covered with a glass or plastic cladding. The cladding has a refractive index lesser than the core to provide total internal reflection to transmit the signal in form of light waves.

Advantages and Disadvantages

- **Advantages of Optical Fibre**

1. **Greater Bandwidth & Faster**
2. **Cheap**—Several miles of optical fibre cable can be made cheaper than equivalent lengths of copper wire.
3. **Thinner and Lightweight—**Optical fibre is thinner, and can be drawn to smaller diameters than copper wire. They are of smaller size and lighter weight.
4. **Higher carrying capacity—**Because optical fibers are much thinner than copper wires, more fibers can be bundled into a given-diameter cable. This allows more phone lines to go over the same cable or more channels to come through the cable into your cable TV box.
5. **Less signal degradation—**The loss of signal in optical fiber is less than that in copper wire.
6. **Light signals—**Unlike electrical signals transmitted in copper wires, light signals from one fiber do not interfere with those of other fibers in the same fiber cable. This means clearer phone conversations or TV reception.
7. **Long Lifespan**—Optical fibers usually have a longer life cycle of over 100 years.

- **Disadvantages of Optical Fiber**

1. **Limited Application—**Fiber optic cable can only be used on ground, and it cannot leave the ground or work with the mobile communication.
2. **Low Power—**Light-emitting sources are limited to low power.
3. **Fragility—**Optical fibre is rather fragile and more vulnerable to damage compared to copper wires.

Wireless Media- Radio waves

Radio waves are easy to generate, can travel long distances, and can penetrate buildings easily, making them widely used for both indoor and outdoor communication.

- **Characteristics & Propagation:** Omnidirectional (travel in all directions from the source). They do not require line-of-sight alignment between the transmitter and receiver. They can pass through solid obstacles like walls and they travel in circular pattern in all directions. Any receiver in that range can intercept the singal.
- **Frequency Range:** 3 KHz to 1 GHz.
- **Applications:** FM/AM radio broadcasting, cellular telephony, Wi-Fi networks, and garage door openers.
- **Advantages:**
    - Wide coverage area due to omnidirectional property.
    - Can penetrate solid structures easily.
- **Disadvantages:**
    - Susceptible to electrical interference and noise.
    - Low security (easy to intercept since signals spread in all directions).

Wireless Media- Microwaves

Microwaves provide high-bandwidth communication links but require strict line-of-sight propagation.

- **Characteristics & Propagation:** Highly directional (propagates in a straight line). Because they cannot bend around obstacles like the curvature of the Earth or large buildings, both the transmitting and receiving antennas must be in direct line-of-sight of each other.
- **Frequency Range:** 1 GHz to 300 GHz
- **Applications:** Terrestrial long-distance telephone communication, mobile phone tower-to-tower links, and radar systems.
- **Advantages:**
  - Supports high data transmission rates (higher bandwidth than radio waves).
  - Does not require expensive physical cables for difficult terrains like mountains or water bodies.
- **Disadvantages:**
  - Line-of-sight constraint requires tall towers (repeater stations every 30–50 km due to Earth's curvature).
  - Affected by weather conditions like heavy rain attenuation.

Wireless Media- Infrared

Infrared waves are short-range, high-frequency waves restricted by physical barriers, making them naturally secure for localized device communication.

- **Characteristics & Propagation:** Used for very short-range communication. They cannot penetrate solid objects (walls, doors), which prevents interference between rooms.
- **Frequency Range:** 300 GHz to 400 THz.
- **Applications:** TV/AC remote controls, wireless computer mice and keyboards, and short-range data transfer (older infrared ports on mobile devices).
- **Advantages:**
  - High security because signals cannot pass through walls.
  - Immune to electrical interference from external radio sources.
- **Disadvantages:**
  - Limited to very short distances (line-of-sight within the same room).
  - Performance degrades in bright sunlight (due to infrared components in sunlight).

Wireless Media- Satellites

Satellite communication is a specialized form of microwave transmission that uses artificial satellites orbiting the Earth to bridge vast geographical distances.

- **Characteristics & Propagation:** Operates on microwave frequencies using an Earth station that transmits a signal (uplink) to a satellite, which amplifies and retransmits the signal back to another Earth station (downlink) using a different frequency.
- **Applications:** Global positioning systems (GPS), Direct-to-Home (DTH) television broadcasting, international telephony, and global weather forecasting.
- **Advantages:**
  - Provides truly global coverage, reaching remote rural areas, oceans, and deserts where laying cables is impossible.
  - Single broadcast can reach millions of receivers simultaneously (Point-to-Multipoint).
- **Disadvantages:**
  - High propagation delay (latency) due to the immense distance signals travel to space and back (~36,000 km for GEO).
  - High setup and maintenance cost.

CONNECTING WIRELESSLY TO INTERNET

WIFI	WIMAX
Wifi is defined under IEEE 802.11x standards where x stands for various WiFi versions.	WiMax is defined under IEEE 802.16y standards where y stands for various WiMax versions.
WiFi is for LAN (Local Area Network) applications.	WiMax is for MAN (Metropolitan Area Network) applications.
WiFi does not guarantee any Quality of Service (Qos).	WiMax guarantee Quality of Service (Qos).
WiFi network range is around 100 meters.	WiMax network can reach about 50-90 km.
WiFi MAC layer uses CSMA/CA protocol which is not connection oriented.	WiMax is connection oriented in nature.
WiFi is short range technology.	WiMax is long range technology.
WiFi connection can transmit up to 54 mbps.	WiMax connection can transmit up to 70 mbps.

Networking Devices(HUBS)

- A hub is basically **a multiport repeater.**
- A hub connects multiple wires coming from different branches.
- Hubs **cannot filter data**, so **data packets are sent to all connected devices**.
- Also, they **do not have intelligence to find out best path** for data packets which leads to inefficiencies and wastage.

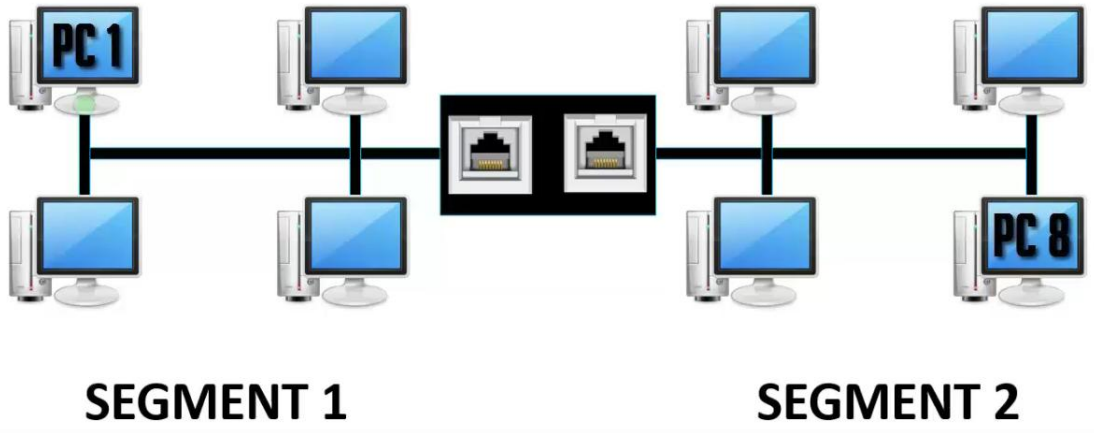


Repeaters

- A repeater operates at the **physical layer.**
- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network.
- An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength.
- It is a **2 port device**.
- Repeaters are used after every 100m.

Networking Devices(Bridges)

- A bridge operates at **data link layer connecting two LANs**.
- A bridge has a functionality of **filtering content by reading the MAC addresses** of source and destination.
- It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a **2 port device**.



Networking Devices(Switches)

- Switch is **data link layer** device.
- Switch can **perform error checking** before forwarding data, that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only.

- **NOTE: Bridges and switches can connect two LANs so used in same building for connection**

Networking Devices(Routers)

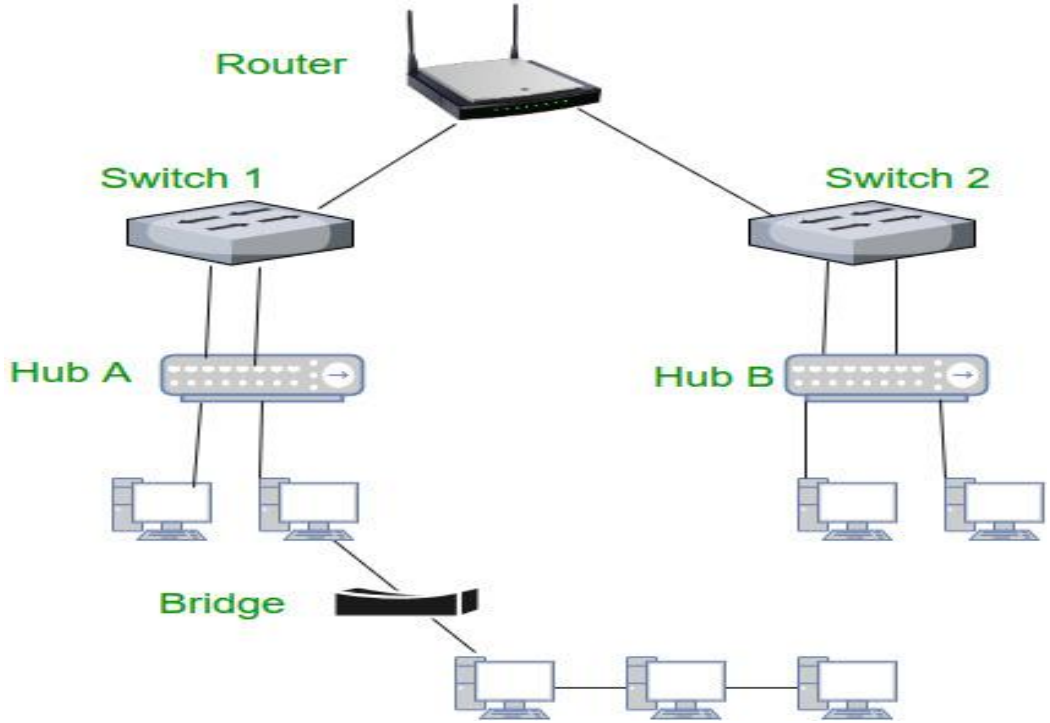
- Router is mainly a **Network Layer device**
- A router is a device like a switch that **routes data packets based on their IP addresses.** Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on the best route for data packets.

Switches



Routers





Networking Devices(Gateways)

- Gateways are also called **protocol converters** and can operate **at any network layer.**
- A gateway, as the name suggests, is a passage to **connect two networks together that may work upon different networking models**.
- They basically works as the messenger agents that take data from one system, interpret it, and transfer it to another system

Star Topology

•Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
•The central computer is known as a **server**,
and the peripheral devices attached to the server are known as **clients**.



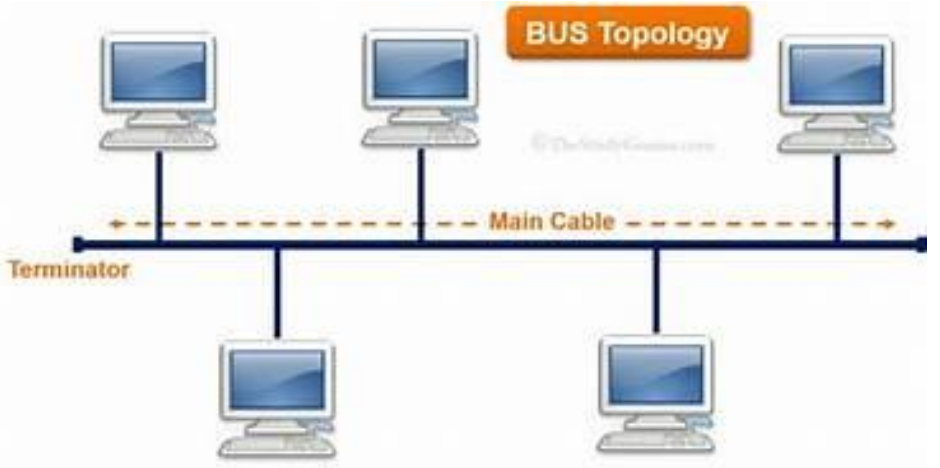
Advantages:

•**Easily expandable:** It is easily expandable as new stations can be added to the open ports on the hub.
•**Cost effective:** Star topology networks are cost-effective as it uses inexpensive coaxial cable.
•**High data speeds:** It supports a bandwidth of approx 100Mbps.

Disadvantages:

•**A Central point of failure:** If the central hub or switch goes down, then all the connected nodes will not be able to communicate with each other.
•**Cable:** Sometimes cable routing becomes difficult when a significant amount of routing is required.

Bus Topology



•The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
•Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
•When a node wants to send a message over the network, it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.

Advantages:

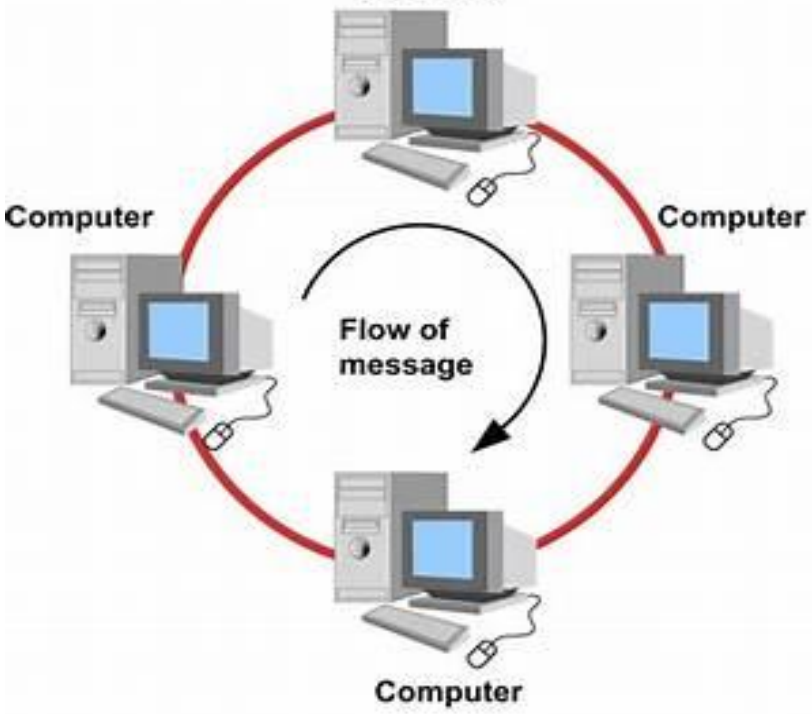
Low-cost cable: In bus topology, nodes are directly connected to the cable without passing through a hub. Therefore, the initial cost of installation is low.
Limited failure: A failure in one node will not have any effect on other nodes.

Disadvantages:

•**Difficult troubleshooting:** If any fault occurs in the cable, then it would disrupt the communication for all the nodes.
•**Signal interference:** If two nodes send the messages simultaneously, then the signals of both the nodes collide with each other.
Reconfiguration difficult: Adding new devices to the network would slow down the network.

Ring Topology

•Ring topology is like a bus topology, but with connected ends.
•The node that receives the message from the previous computer will retransmit to the next node.
•It has no terminated ends, i.e., each node is connected to other node and having no termination point.
•The data in a ring topology flow in a clockwise direction.
•The most common access method of the ring topology is **token passing**.



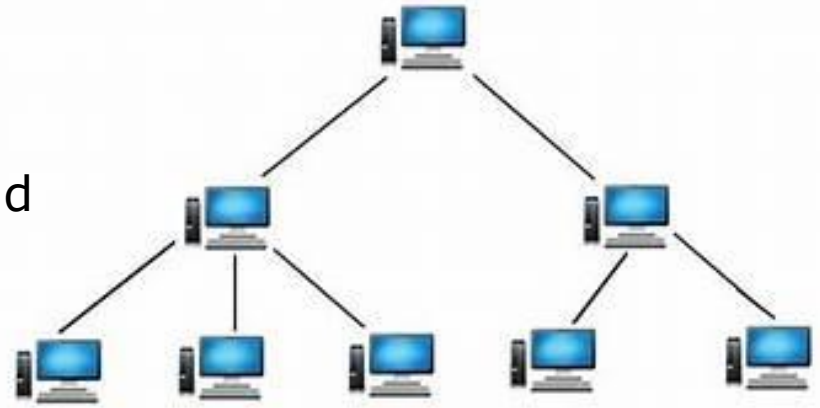
Advantages:

•**Network Management:** Faulty devices can be removed from the network without bringing the network down.
•**Cost:** Twisted pair cabling is inexpensive and easily available. Therefore, the installation cost is very low.
•**Reliable:** It is a more reliable network because the communication system is not dependent on the single host computer.

Disadvantages:

•**Difficult troubleshooting:** If any fault occurs in the cable, then it would disrupt the communication for all the nodes.
•**Failure:** The breakdown in one station leads to the failure of the overall network.
•**Reconfiguration difficult:** Adding new devices to the network would slow down the network.

Tree Topology



- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion.
- The top-most node in tree topology is known as a root node, and all other nodes are the descendants of the root node.
- There is only one path exists between two nodes for the data transmission. Thus, it forms a parent-child hierarchy

Advantages:

- **Easily expandable:** We can add the new device to the existing network. Therefore, we can say that tree topology is easily expandable.
- **Easy Error detection**
- **Limited failure:** The breakdown in one station does not affect the entire network.

Disadvantages:

- **Difficult troubleshooting:** If any fault occurs in the node, then it becomes difficult to troubleshoot the problem.
- **Failure:** A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.
- **Reconfiguration difficult:** If new devices are added, then it becomes difficult to reconfigure.

Mesh Topology

- Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.
- The Internet is an example of the mesh topology.
- Mesh topology can be formed by using the formula:

Number of cables = (n*(n-1))/2;

